Voltage-Current Protection System

Design Approach: The main aim of this project is to make a microcontroller based system that disconnects when over voltage or over current are detected. I used Arduino Uno as the main microcontroller which mainly monitors everything and provides information in every connecting pin based on the firmware. Two 10KΩ potentiometers for mannualy changes voltage (0-400)V and current(0-14)A to test the whole system. A single channel relay with LED for the Load status and conditional operation. One LCD for visualizing status and overall and one Keypad for changing the features of the system.

Implemetation: The whole system is built in hardware level and also simulated in cirkit online simulator. Two 10KΩ potentiometers are connected to Arduino Uno. Two legs of both potentiometer are connected to 5V and Gnd and wiper of potentiometer 1 is connected to A0 showcasing the voltage regulation and wiper of potentiometer 2 is connected to A1 for current regulation. A0 and A1 read analog values and convert that in digital values. So, for this conversion and showcasing voltage and current as real values, I use potentiometer knob range from min(128) and max(895) and set the voltage and current max range (400V & 14A), Based on these fixed parameters simulated voltage and current are produced following these equations as like real voltage and current:

$$V_{sim} = \frac{895 - A0}{767} \times 400 \qquad I_{sim} = \frac{895 - A1}{767} \times 14$$

A LCD displays the reading of voltage and current and their status, There are two status I have used, One is normal, another is Fault,when both voltage and current are lower than the threshold (By default threshold values are: 200V and 10A) Fault status is triggered. A 4x4 keypd is used to chaging this threshold values, clicking “A” from the keypad, give access to user to manipulate, int, float both values can be accepted. “C” for cancel, “D” for deleting the previous digit and “#” to save. A 220Ω resistor with LED is connected via single channel relay. Anode of the LED is connected with D9 with arduino and cathode is connected with common ground through resistor, and most importantly NC is connected with 5V supply, which keeps the LED always ON when the system is in Normal Condition otherwise it will cut of the circuit.

Results: The whole hardware system and simulation both are designed to demonstrate normal and Fault(Over Voltage and Over Current) status. With the adjusting knobs of potentiometers, when the simulated voltage exceeds the limit, the the system successfully triggers the LED based on status and commands the relay output to disconnect the load. This approach fulfilling the core objectives without using dangerous main volts.

Limitations: The core limitation of this system are simulated input, simpulates inputs are mainly used for showcasing the actual behaviour and testing but compared to real time project, voltage and current sensor implementation will be better for low-voltage appliances. For the simulation purpose and manually changing the voltage and current to showcase their behaviour, the simulated inputs from potentiometers are used.